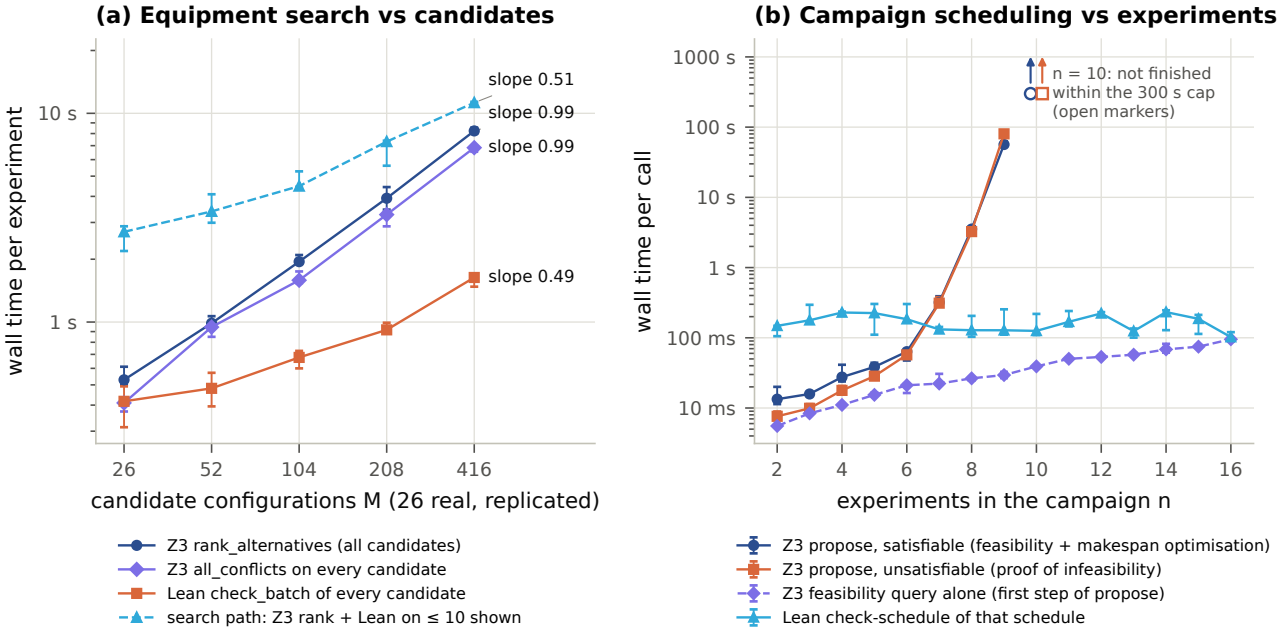
(a) Equipment search vs candidates
wall time per experiment
10 s
1 s
slope 0.51
slope 0.99
slope 0.99
slope 0.49
26
52
104
208
416
candidate configurations M (26 real, replicated)
Z3 rank_alternatives (all candidates)
Z3 all_conflicts on every candidate
Lean check_batch of every candidate
search path: Z3 rank + Lean on ≤ 10 shown
(b) Campaign scheduling vs experiments
wall time per call
1000 s
100 s
10 s
1 s
100 ms
10 ms
n = 10: not finished
within the 300 s cap
(open markers)
2
4
6
8
10
12
14
16
experiments in the campaign n
Z3 propose, satisfiable (feasibility + makespan optimisation)
Z3 propose, unsatisfiable (proof of infeasibility)
Z3 feasibility query alone (first step of propose)
Lean check-schedule of that schedule